

Faizaan Rehman

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TECHNICAL SKILLS

Languages: TypeScript; JavaScript; Java; Python; C/C++; SQL; Kotlin; Groovy

Frameworks: React; Angular; Next.js; Node.js; Django; Tailwind; Unreal Engine (UMG)

Tools/AI: Git; Bash; Docker; Android Studio; Linux; Hugging Face; Meta Llama 3.1 8B

EXPERIENCE

Programmer

Sep 2022 – Present

Behaviour Interactive

- Implemented scalable UI systems in **Unreal Engine** for a live-service game with **40k+ concurrent players**
- Optimized friends list rendering (**2,000 entries**) by batching updates, reducing frame stalls from **13.5s to 266ms** and single-entry updates from **10s to 4.7ms**
- Modernized UI by converting **Scaleform** widgets to Unreal Motion Graphics (**UMG**), contributing to a **~250MB memory reduction** across the main menu/HUD
- Designed a templated badge rendering framework supporting **90+ characters**, reducing required UI widgets from **360 character-specific** instances to just **4 reusable templates**
- Improved progression system responsiveness by eliminating input delays and automating chained node collection, reducing user idle times by up to **860ms** per interaction
- Optimized HUD runtime performance by recycling card widgets instead of reconstructing them, reducing allocations from **5+ widgets** per match to **3 reusable instances**

Compiler Engineer

Jan 2022 – Apr 2022

Huawei Technologies Canada

- Extended our Gem5 C++ processor model to handle premature loop exits, improving branch prediction accuracy

Software Developer

May 2020 – Apr 2021

HubHead Corp.

- Developed **Groovy** automation scripts to migrate **hundreds** of legacy desktop records to cloud systems via **REST APIs**, eliminating manual migration steps
- Used **AngularJS** and **HTML/CSS** to build interactive web components resulting in an intuitive UI for clients

PROJECTS

Restaurant Deal Aggregator | *React, Next.js, Node.js, TypeScript, PostgreSQL*

Apr 2025 – Present

- Architected a **rate-limited** scraping pipeline using **Regex** to extract time-sensitive deals and date ranges
- Improved deal detection precision from **52% to 76%** by refining classification logic on **200 samples**
- Integrated **LLM inference** via **Hugging Face** using **Llama 3.1 8B** with token-efficient prompts to generate concise deal titles, keeping API costs below **\$0.10/month**

Student Housing Website | *React, Django, Python, PostgreSQL, Firebase, Google Drive*

May 2022 – Apr 2023

- Built a cloud-based file storage pipeline using the **Google Drive API**, storing media externally and persisting only URLs in the database to reduce bloat
- Designed a real-time chat system using **Firebase Cloud Firestore**, enabling live messaging between students and property owners without requiring a custom WebSocket backend

Food Expiry Tracker | *Kotlin, Android Studio, Google Maps*

Jan 2023 - Apr 2023

- Implemented background notifications with **Android WorkManager** and **NotificationManager** to alert users of ingredients nearing expiration
- Facilitated location-based food bank discovery via **Google Maps Intents**, streamlining food donation for users

EDUCATION

University of Waterloo

Waterloo, ON

B.A.Sc. in Computer Engineering, GPA: 3.8

Sep 2018 – Apr 2023

- *Distinction; Dean's Honour List*